

You have 2 hours for completing the exam.

No electronic devices are allowed (i.e. calculators, mobile phones, etc.).

All answers must include the supporting deductions and calculations (if the calculations are too heavy then they can be left implied in the final answer).

If some answer does not include all the necessary information, it will be deemed invalid.

Name and surname:

1. An idle bookkeeper is waiting for clients, and in the meantime he spends his time playing around with the word 'BOOKKEEPER'. He's trying to make a list of all the distinct words that can be constructed with the letters of the word 'BOOKKEEPER', possibly with some additional conditions, but he is not good at Math. Help him find:
 - (a) (1 point) The number of all distinct words where all the vowels come before all the consonants.
 - (b) (1 point) The number of all distinct words where equal letters are always together.
 - (c) (1 point) The number of all distinct words where the two Os are always apart and the two Ks are always apart.
2. (2 points) Prove the following version of the hockey-stick identity:

$$\sum_{j=k}^n \binom{j}{k} = \binom{n+1}{k+1}$$

Hint: Use induction on n while keeping k fixed.

3. Let a_n be the number of n -letter words from the alphabet $\{A, B, C\}$ such that any nonterminal A has to be immediately followed by a B .
 - (a) (1 point) Find a recurrence relation for $\langle a_n \rangle$ (including its initial values).
 - (b) (1 point) Prove that $\langle a_n \rangle$ is monotonically increasing. **Hint:** Use induction on n .
 - (c) (1 point) Solve the recurrence.
4. (2 points) Solve the divide-and-conquer recurrence

$$T_n = \begin{cases} 0 & \text{if } n = 1, \\ 4T_{\frac{n}{2}} + n^2 & \text{for } n = 2^k, k \geq 1. \end{cases}$$

SOME USEFUL TABLES AND FORMULAS

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \text{for } 0 \leq k \leq n$$

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$$

Table of factorials

n	$n!$	n	$n!$	n	$n!$
0	1	1	1	2	2
3	6	4	24	5	120
6	720	7	5 040	8	40 320
9	362 880	10	3 628 800	11	39 916 800

Table of subfactorials

n	$!n$	n	$!n$	n	$!n$
1	0	2	1	3	2
4	9	5	44	6	265
7	1 854	8	14 833	9	133 496
10	1 334 961	11	14 684 570	12	176 214 841

Table of annihilators

Sequence	Annihilator
$\langle c \rangle$	$\mathbb{E} - 1$
$\langle \text{poly. in } n \text{ of degree } k \rangle$	$(\mathbb{E} - 1)^{k+1}$
$\langle c^n \rangle$	$\mathbb{E} - c$
$\langle c^n \times \text{poly. in } n \text{ of degree } k \rangle$	$(\mathbb{E} - c)^{k+1}$